CEFET/RJ

DAL & Comunidade

APPLY

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Implicitly Loops – sapply, lapply

- lapply, sapply executa uma função para cada coluna
  - O primeiro caractere define o tipo de retorno
    - l – lista, s – simples (vetor ou matriz)
  - O segundo parâmetro é a função a ser invocada
  - Os seguintes parâmetros são passados para a função invocada
- apply é a função genérica
  - O segundo parâmetro define:
    - se chama a função para cada linha (1) ou cada coluna (2)

Exemplo

```r
weight <- c(60, 72, 57, 90, 95, 72)
height <- c(1.75, 1.80, 1.65, 1.90, 1.74, 1.91)
subject <- c("A", "B", "C", "D", "E", "F")
```

Criando uma tabela básica

```
d <- data.frame(weight=weight, height=height, subject=subject)
head(d)
```

A data.frame: 6 × 3

	weight	height	subject
	<dbl>	<dbl>	<chr>
1	60	1.75	A
2	72	1.80	B
3	57	1.65	C
4	90	1.90	D
5	95	1.74	E
6	72	1.91	F

Implicitly Loops – sapply, lapply

```
lapply(d[,1:2], mean, na.rm=TRUE)
```

$weight
74.3333333333333
$height
1.79166666666667

```
sapply(d[,1:2], mean, na.rm=TRUE)
```

weight: 74.3333333333333 **height:** 1.79166666666667

lapply and sapply process each column invoking the function passed as second parameter.
All the following parameters are passed to the function.

Implicitly Loops – sapply, lapply

```
m <- as.matrix(d[,1:2])
```

```
apply(m, 1, min, na.rm=TRUE)
```

1.75 · 1.8 · 1.65 · 1.9 · 1.74 · 1.91

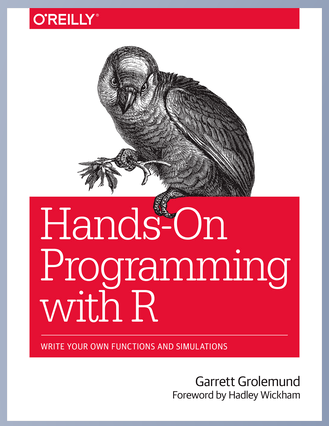
```
apply(m, 2, min, na.rm=TRUE)
```

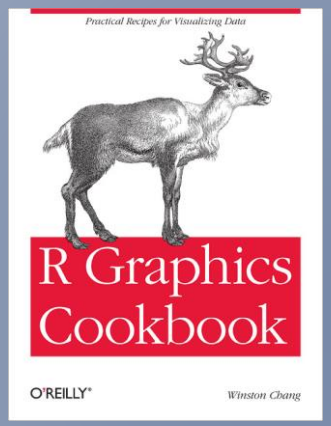
weight: 57 **height:** 1.65

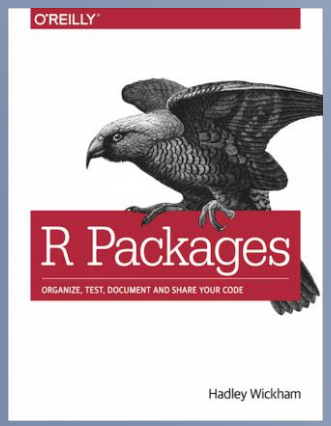
The first parameter of apply is a matrix.
The second parameter of apply is the direction (1: row, 2: column)

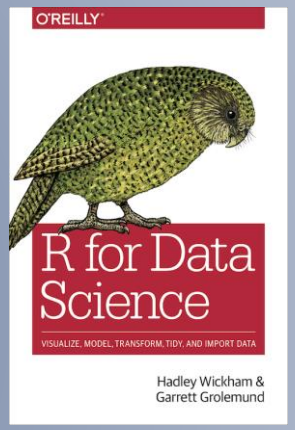
Referências

Código: https://nbviewer.org/github/eogasawara/R/blob/main/11-Apply.ipynb







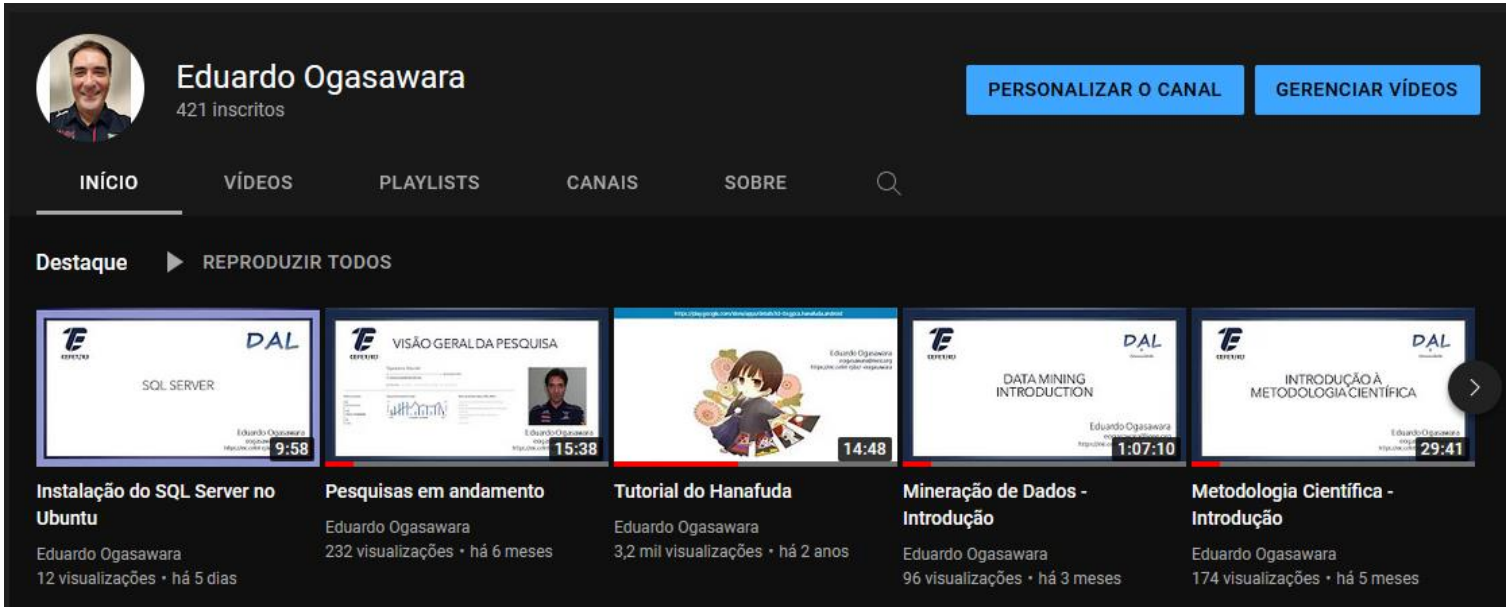


Hands-on Programming with R: https://rstudio-education.github.io/hopr/index.html
R Graphics Cookbook: https://r-graphics.org
R Packages: https://r-pkgs.org/index.html
R for Data Science: https://r4ds.had.co.nz

[1] G. Grolemund, 2014, Hands-On Programming with R: Write Your Own Functions and Simulations. O'Reilly Media, Inc.

Mantenha-se atualizado

Inscreva-se: https://eic.cefet-rj.br/~eogasawara/youtube



Slides and Jupyter notebooks: https://github.com/eogasawara/R